Huijun Jin

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Ph.D. expected February 2027. Available for a postdoctoral position from March 2027.

Research Interests

Database systems and vector data management: graph-based approximate nearest neighbor search (HNSW), index maintenance under updates, robustness to out-of-distribution queries, hardware-conscious index layouts, retrieval-augmented generation, and machine learning for database tuning.

Education

Ph.D. in Computer Science Mar. 2024 – Feb. 2027 (expected)
Yonsei University, Seoul, Korea
Dissertation: *Optimizing Efficiency, Updatability, and Robustness of HNSW Indexes for High-Performance Vector Search*
Advisor: Prof. Sanghyun Park

M.S. in Computer Science Mar. 2020 – Feb. 2022
Yonsei University, Seoul, Korea
Advisor: Prof. Sanghyun Park

B.Eng. in Computer Science and Technology Sep. 2014 – Jun. 2018
Yanbian University, Yanji, China

Publications

Selected publications (first author)

[1] **Huijun Jin**, Sanghyun Park. CAMP-HNSW: Hardware-Topological Co-Design for OOD-Robust ANNS via Cache Alignment and Targeted Rewiring. *Proceedings of the VLDB Endowment (PVLDB)*, 20(2):438–451, 2026. To be presented at VLDB 2027. [code]

[2] **Huijun Jin**, Jieun Lee, Shengmin Piao, Sangmin Seo, Sanghyun Park. PRO-HNSW: Proactive Repair and Optimization for High-Performance Dynamic HNSW Indexes. In *Proc. IEEE International Conference on Data Engineering (ICDE)*, 2026. [code]

[3] **Huijun Jin**, Jieun Lee, Shengmin Piao, Sangmin Seo, Sein Kwon, Sanghyun Park. Efficient Approximate Nearest Neighbor Search via Data-Adaptive Parameter Adjustment in Hierarchical Navigable Small Graphs. In *Proc. Design, Automation & Test in Europe Conference (DATE)*, 2025. [code]

Other peer-reviewed publications

[4] Sein Kwon, Youngwan Jo, Seungyeon Choi, Jieun Lee, **Huijun Jin**, Sanghyun Park. LatentTune: Efficient Tuning of High Dimensional Database Parameters via Latent Representation Learning. In *Proc. IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC)*, 2025.

[5] Jieun Lee, Sangmin Seo, Chanho Yeom, **Huijun Jin**, Sein Kwon, Sanghyun Park. An Efficient and Noise-Robust Framework for High-Dimensional Tuning in Big Data Analytics. *Engineering Applications of Artificial Intelligence*, 158:111332, 2025.

[6] **Huijun Jin**, Jieun Lee, Sanghyun Park. RTune: A RocksDB Tuning System with Deep Genetic Algorithm. In *Proc. Genetic and Evolutionary Computation Conference (GECCO)*, 2022.

[7] **Huijun Jin**, Won Gi Choi, Jonghwan Choi, Hanseung Sung, Sanghyun Park. Improvement of RocksDB Performance via Large-Scale Parameter Analysis and Optimization. *Journal of Information Processing Systems*, 18(3):374–388, 2022.

Publications in Korean venues

Seven papers in Korean venues (Communications of the KIISE, KIISE Transactions on Computing Practices, KCC, KIPS), including a first-author survey on retrieval-augmented generation and vector databases in a special issue of Communications of the KIISE (2024). Full list at huijunjin.github.io/publications.

Patents

[P1] Sanghyun Park, **Huijun Jin**, Jieun Lee, Shengmin Piao, Sangmin Seo, Sein Kwon. Method and Program for Approximate Nearest Neighbor Search via Data-Adaptive Parameter Adjustment. U.S. Patent Application Publication US 2026/0141002 A1, May 2026 (priority: Korean Patent Application 10-2024-0166800, Nov. 2024).

[P2] Sanghyun Park, **Huijun Jin**, Jieun Lee. Database Tuning Method Using Deep Genetic Algorithm. Korean Patent No. 10-2727561, granted Nov. 2024.

[P3] Sanghyun Park, **Huijun Jin**, Jonghwan Choi, Won Gi Choi, Hanseung Sung. Method and Apparatus for Tuning Database Based on Machine Learning. Korean Patent No. 10-2703138, granted Sep. 2024.

Also co-inventor of two pending Korean patent applications in database parameter tuning.

Research Experience

Participant, S/W Starlab project (IITP, Korea) Mar. 2020 – Aug. 2022, Mar. 2024 – Dec. 2024
Government-funded project on a high-performance in-memory distributed DBMS based on flash memory storage (IITP-2017-0-00477), during M.S. and Ph.D. studies.

Researcher (post-master's), Data Engineering Lab, Yonsei University Mar. 2022 – Aug. 2022
Software development and database testing for the S/W Starlab project.

Conference Presentations

- CAMP-HNSW — VLDB 2027, Athens, Greece (scheduled, Aug. 2027)
- PRO-HNSW — ICDE 2026, Montréal, Canada
- DHNSW — DATE 2025, Lyon, France
- RTune — GECCO 2022, Boston, USA

Teaching

Teaching Assistant, *Software Capstone Design*, Yonsei University Sep. 2020 – Dec. 2020
Mentored one undergraduate capstone project team.

Teaching Assistant, *Database Systems*, Yonsei University Mar. 2020 – Jun. 2020

Teaching Assistant, Yanbian University Jul. 2018 – Jun. 2019
Assisted undergraduate courses across the department: in-class interpretation and help with assignments.

Awards and Scholarships

- Best Paper Award, KIPS Fall Conference (Korea Information Processing Society) 2020
- Academic Research Fellowship (ARF), Yonsei University 2026
- BK21 Scholarship, Yonsei University 2025 – 2026
- Special Scholarship for International Students, Yonsei University 2020 – 2022, 2024 – 2025
- National Scholarship, China 2017

Skills and Languages

Programming C++ (index engines), Python (experiment pipelines and analysis)
Systems Graph-based vector indexes (HNSW), RocksDB, Redis
Languages Korean (native), Chinese (native), English (professional working proficiency)

References

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